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DS Homework Set – 1

1) Sum of two matrices

```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;
    printf("Enter the number of rows and columns: ");
    scanf("%d%d", &r, &c);
    printf("Enter the elements of first matrix:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    printf("Enter the elements of second matrix:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }
    printf("Sum of the matrices:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            printf("%d ", sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output :

Enter the number of rows and columns: 2 2

Enter the elements of first matrix:

1 2

3 4
Enter the elements of second matrix:
5 6
7 8
Sum of the matrices:
6 8
10 12

2) Write a C program to read n numbers into an array and display them in reverse order.

```
#include <stdio.h>
int main()
{
    int a[100], n, i;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("Elements in reverse order:\n");
    for(i = n - 1; i >= 0; i--)
    {
        printf("%d ", a[i]);
    }
    return 0;
}
```

Output :

Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Elements in reverse order:
50 40 30 20 10

3) Write a C program to count the total number of duplicate elements in an array.

```
#include <stdio.h>
int main()
{
    int a[100], n, i, j, count = 0;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter the array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
```

```

        if(a[i] == a[j])
        {
            count++;
            break;
        }
    }
}
printf("Total number of duplicate elements = %d", count);
return 0;
}

```

Output :

Enter the number of elements: 7
Enter the array elements:
10 20 10 30 20 40 10
Total number of duplicate elements = 2

4) Write a C program to print all unique elements in an array.

```

#include <stdio.h>
int main()
{
    int a[100], n, i, j, count;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter the array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("Unique elements are:\n");
    for(i = 0; i < n; i++)
    {
        count = 0;
        for(j = 0; j < n; j++)
        {
            if(a[i] == a[j])
            {
                count++;
            }
        }
        if(count == 1)
        {
            printf("%d ", a[i]);
        }
    }
    return 0;
}

```

Output :

Enter the number of elements: 7
Enter the array elements:
10 20 10 30 20 40 50

Unique elements are:

30 40 50

5) Write a C program to insert an element in a sorted array.

```
#include <stdio.h>
int main()
{
    int a[100], n, i, element, pos;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter the sorted array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("Enter the element to insert: ");
    scanf("%d", &element);
    for(i = 0; i < n; i++)
    {
        if(a[i] > element)
        {
            pos = i;
            break;
        }
    }
    if(i == n)
        pos = n;
    for(i = n; i > pos; i--)
    {
        a[i] = a[i - 1];
    }
    a[pos] = element;
    n++;
    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    return 0;
}
```

Output :

Enter the number of elements: 5

Enter the sorted array elements:

10 20 30 40 50

Enter the element to insert: 35

Array after insertion:

10 20 30 35 40 50

6) Write a C program to insert an element in an unsorted array.

```
#include <stdio.h>
```

```

int main()
{
    int a[100], n, i, element, pos;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    printf("Enter the array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }
    printf("Enter the position to insert (1 to %d): ", n + 1);
    scanf("%d", &pos);
    printf("Enter the element to insert: ");
    scanf("%d", &element);
    for(i = n; i >= pos; i--)
    {
        a[i] = a[i - 1];
    }
    a[pos - 1] = element;
    n++;
    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }
    return 0;
}

```

Output :

```

Enter the number of elements: 5
Enter the array elements:
10 20 30 40 50
Enter the position to insert (1 to 6): 3
Enter the element to insert: 25
Array after insertion:
10 20 25 30 40 50

```

7) Write a C program for Matrix Multiplication.

```

#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2;
    int i, j, k;
    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d", &r1, &c1);
    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d", &r2, &c2);
    if(c1 != r2)
    {
        printf("Matrix multiplication is not possible.");
        return 0;
    }
}

```

```

}
printf("Enter elements of first matrix:\n");
for(i = 0; i < r1; i++)
{
    for(j = 0; j < c1; j++)
    {
        scanf("%d", &a[i][j]);
    }
}
printf("Enter elements of second matrix:\n");
for(i = 0; i < r2; i++)
{
    for(j = 0; j < c2; j++)
    {
        scanf("%d", &b[i][j]);
    }
}
for(i = 0; i < r1; i++)
{
    for(j = 0; j < c2; j++)
    {
        c[i][j] = 0;
        for(k = 0; k < c1; k++)
        {
            c[i][j] += a[i][k] * b[k][j];
        }
    }
}
printf("Product of the matrices:\n");
for(i = 0; i < r1; i++)
{
    for(j = 0; j < c2; j++)
    {
        printf("%d ", c[i][j]);
    }
    printf("\n");
}
return 0;
}

```

Output :

```

Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter elements of first matrix:
1 2
3 4
Enter elements of second matrix:
5 6
7 8
Product of the matrices:
19 22
43 50

```

8) Write a C program to find the Transpose of a Matrix.

```
#include <stdio.h>
int main()
{
    int a[10][10], t[10][10];
    int r, c, i, j;
    printf("Enter the number of rows and columns: ");
    scanf("%d%d", &r, &c);
    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            t[j][i] = a[i][j];
        }
    }
    printf("Transpose of the matrix:\n");
    for(i = 0; i < c; i++)
    {
        for(j = 0; j < r; j++)
        {
            printf("%d ", t[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output :

Enter the number of rows and columns: 2 3

Enter the elements of the matrix:

1 2 3

4 5 6

Transpose of the matrix:

1 4

2 5

3 6

9) Write a C program to find the sum of the right diagonal (secondary diagonal) of a matrix.

```
#include <stdio.h>
int main()
{
    int a[10][10];
    int n, i, j, sum = 0;
```

```

printf("Enter the order of the square matrix: ");
scanf("%d", &n);
printf("Enter the elements of the matrix:\n");
for(i = 0; i < n; i++)
{
    for(j = 0; j < n; j++)
    {
        scanf("%d", &a[i][j]);
    }
}
for(i = 0; i < n; i++)
{
    sum = sum + a[i][n - 1 - i];
}
printf("Sum of the right diagonal = %d", sum);
return 0;
}

```

Output :

```

Enter the order of the square matrix: 3
Enter the elements of the matrix:
1 2 3
4 5 6
7 8 9
Sum of the right diagonal = 15

```

10) Write a C program to check whether a matrix is an Identity Matrix.

```

#include <stdio.h>
int main()
{
    int a[10][10];
    int n, i, j, flag = 1;
    printf("Enter the order of the square matrix: ");
    scanf("%d", &n);
    printf("Enter the elements of the matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j)
            {
                if(a[i][j] != 1)
                {
                    flag = 0;
                }
            }
        }
    }
}

```



```

    }
    else
    {
        if(a[i][j] != 0)
        {
            flag = 0;
        }
    }
}
}
if(flag == 1)
    printf("The matrix is an Identity Matrix.");
else
    printf("The matrix is not an Identity Matrix.");
return 0;
}

```

Output :

Enter the order of the square matrix: 3

Enter the elements of the matrix:

1 0 0

0 1 0

0 0 1

The matrix is an Identity Matrix.

11) Write a C program to count the number of vowels, consonants, digits, and white spaces in a string.

```

#include <stdio.h>
int main()
{
    char str[100];
    int i;
    int vowels = 0, consonants = 0, digits = 0, spaces = 0;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'A' && str[i] <= 'Z') || (str[i] >= 'a' && str[i] <= 'z'))
        {
            if(str[i] == 'A' || str[i] == 'E' || str[i] == 'I' || str[i] == 'O' || str[i] == 'U' ||
               str[i] == 'a' || str[i] == 'e' || str[i] == 'i' || str[i] == 'o' || str[i] == 'u')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
    }
}

```

```

    else if(str[i] == ' ')
    {
        spaces++;
    }
}
printf("Vowels = %d\n", vowels);
printf("Consonants = %d\n", consonants);
printf("Digits = %d\n", digits);
printf("White spaces = %d\n", spaces);
return 0;
}

```

Output :

Enter a string: Hello World 123
Vowels = 3
Consonants = 7
Digits = 3
White spaces = 2

12) Write a C program to find the frequency of each character in a string.

```

#include <stdio.h>
int main()
{
    char str[100];
    int i, j, count;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        count = 1;
        if(str[i] == '0')
            continue;
        for(j = i + 1; str[j] != '\0'; j++)
        {
            if(str[i] == str[j])
            {
                count++;
                str[j] = '0';
            }
        }
        if(str[i] != '0' && str[i] != '\n')
        {
            printf("%c = %d\n", str[i], count);
        }
    }
    return 0;
}

```

Output :

Enter a string: hello world
h = 1
e = 1

```
l = 3
o = 2
  = 1
w = 1
r = 1
d = 1
```

13) Write a C program to count the number of words in a string.

```
#include <stdio.h>
int main()
{
    char str[100];
    int i, words = 0;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        if((i == 0 && str[i] != ' ' && str[i] != '\n') ||
            (str[i] != ' ' && str[i] != '\n' && str[i - 1] == ' '))
        {
            words++;
        }
    }
    printf("Number of words = %d", words);
    return 0;
}
```

Output :

```
Enter a string: C programming is easy
Number of words = 4
```

14) Write a C program to sort an array of strings in alphabetical order.

```
#include <stdio.h>
#include <string.h>
int main()
{
    char str[20][100], temp[100];
    int n, i, j;
    printf("Enter the number of strings: ");
    scanf("%d", &n);
    getchar();
    printf("Enter the strings:\n");
    for(i = 0; i < n; i++)
    {
        fgets(str[i], sizeof(str[i]), stdin);
        str[i][strcspn(str[i], "\n")] = '\0';
    }
    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
```

```

        if(strcmp(str[i], str[j]) > 0)
        {
            strcpy(temp, str[i]);
            strcpy(str[i], str[j]);
            strcpy(str[j], temp);
        }
    }
}
printf("Strings in alphabetical order:\n");
for(i = 0; i < n; i++)
{
    printf("%s\n", str[i]);
}
return 0;
}

```

Output :

Enter the number of strings: 5

Enter the strings:

banana

apple

mango

grape

orange

Strings in alphabetical order:

apple

banana

grape

mango

orange

15) Write a C program to toggle the case of each character in a sentence (uppercase to lowercase and lowercase to uppercase).

```

#include <stdio.h>
int main()
{
    char str[100];
    int i;
    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
        else if(str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
    }
    printf("After toggling the case:\n%s", str);
}

```

```
    return 0;  
}
```

Output :

Enter a sentence: Hello World

After toggling the case:

hELLO wORLD